Thesis Exposé

Transfer or Relearning? Factual Knowledge in Turkish-Adapted Language Models

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Abstract

This thesis investigates whether improved factual knowledge retrieval in Turkish after Turkish adaptation reflects cross-lingual transfer of previously acquired English parametric knowledge or reaffirmation and relearning from Turkish adaptation data. To address this question, the study proposes a controlled synthetic-fact setup in which fictitious facts are first taught in English and then evaluated after Turkish adaptation under different conditions. In particular, the design compares adaptation without Turkish repetition of the target facts against adaptation with Turkish repetition, making it possible to distinguish transfer from Turkish-side reaffirmation or relearning more directly. The framework also allows controlled analysis of factors such as fact frequency and name distribution while minimizing contamination from naturally occurring facts. The thesis aims to clarify what target-language adaptation changes in multilingual factual access.

1 Introduction and Motivation

Large language models do not retrieve factual knowledge uniformly across languages. Schott et al., 2023 show broad multilingual limitations in encyclopedic factual recall, while Qi et al., 2025 demonstrate that semantically equivalent prompts can yield inconsistent answers across languages. Aggarwal et al., 2025 further show that factuality depends strongly on the language of inquiry, and Chua et al., 2025 argue that these failures reflect broader cross-lingual knowledge barriers rather than merely surface-level multilingual ability. M. Wang et al., 2025 provide a mechanistic perspective by showing that multilingual models may encode knowledge in partially language-independent internal spaces but fail when mapping that knowledge to the target-language output space. Taken together, these findings suggest that multilingual factual access is often language-sensitive rather than uniformly multilingual.

This issue is especially important in the context of target-language adaptation. When a model is adapted toward Turkish and later becomes able to answer factual questions in Turkish, it is not immediately clear what has changed inside the model. One possibility is **cross-lingual transfer**: the model had already acquired the relevant fact in English, and Turkish adaptation merely enabled access to that existing parametric knowledge in Turkish. Another possibility is **relearning**: the fact was effectively learned again from Turkish adaptation data. Distinguishing between these possibilities matters scientifically because it bears on how multilingual knowledge is represented and accessed in adapted language models. It also matters methodologically, because gains in Turkish factual retrieval could otherwise be misinterpreted as evidence of transfer when they may instead reflect Turkish-side reaffirmation or relearning.

This thesis addresses that distinction directly. It asks whether factual knowledge retrieved in Turkish after Turkish adaptation reflects transfer from previously acquired English knowledge or whether it depends on reaffirmation and relearning from Turkish adaptation data. To make this question testable, the thesis proposes a controlled synthetic-fact setup in which fictitious facts are first taught in English and then evaluated after Turkish adaptation under different adaptation conditions.

2 Related Work

Related work can be grouped into four strands: multilingual factual probing and consistency, mechanistic studies of multilingual factual knowledge, target-language adaptation for low-resource languages, and controlled factual intervention through editing or synthetic data. Together, these lines of work show that multilingual factual retrieval is important but fragile. However, they do not directly test whether improved factual retrieval in Turkish after adaptation reflects transfer of previously acquired English knowledge or Turkish-side relearning.

A. Multilingual factual probing and consistency

A first line of work shows that factual access is not uniform across languages. Multilingual evaluation studies report substantial variation in factual recall across languages Schott et al., 2023, limited consistency across semantically equivalent prompts Qi et al., 2025, and strong language-of-inquiry effects Aggarwal et al., 2025. Related work further argues that these patterns reflect broader cross-lingual knowledge barriers rather than merely surface-level multilingual ability Chua et al., 2025; Kang and Kim, 2025. However, these studies do not examine target-language adaptation as the intervention.

B. Mechanisms of multilingual factual knowledge

A second line of work investigates how multilingual factual knowledge is represented and acquired. Prior studies distinguish between independent, shared, and transferred factual knowledge Zhao et al., 2024, trace multilingual factual acquisition during pretraining Liu et al., 2025, and show that cross-lingual inconsistency can arise when internally represented knowledge is not successfully mapped into the target-language output space M. Wang et al., 2025. These studies help explain why multilingual factual retrieval may fail, but they do not test whether Turkish adaptation changes access through transfer or relearning.

C. Target-language adaptation and forgetting

A third line of work studies adaptation to low-resource or target languages, including Turkish, and often focuses on performance gains and catastrophic forgetting. Work on Turkish adaptation and multilingual continual learning shows that target-language adaptation can improve performance while also risking degradation of previously acquired capabilities Acikgoz et al., 2024; Winata et al., 2023; Yamaguchi et al., 2026; Zhao et al., 2025. However, these studies do not isolate the source of improved factual retrieval at the level of controlled facts.

D. Controlled interventions and synthetic settings

The closest methodological neighbors to the present thesis are cross-lingual knowledge editing and synthetic factual benchmarks. Cross-lingual editing studies test whether fact-specific updates propagate across languages, but they manipulate individual facts directly rather than examining target-language adaptation J. Wang et al., 2024. Synthetic benchmarks such as TOFU show the value of fictitious data for controlling knowledge source and contamination Maini et al., 2024. To the best of my knowledge, however, no prior work combines controlled synthetic facts with Turkish adaptation in order to compare adaptation without Turkish fact repetition against adaptation with Turkish fact repetition.

3 Research Gap

The literature reviewed above establishes three important points: multilingual factual retrieval is often inconsistent across languages, multilingual factual knowledge may be only partially shared across languages, and target-language adaptation can substantially change model behavior. However, these findings still leave the central question of this thesis unresolved. Existing work does not determine whether improved factual retrieval in Turkish after Turkish adaptation reflects access to previously acquired English knowledge or renewed learning from Turkish adaptation data.

To the best of my knowledge, no prior study has combined a controlled synthetic-fact setup with target-language adaptation in order to compare Turkish adaptation *without* repetition of the target facts against Turkish adaptation *with* repetition of those facts. As a result, the literature does not yet provide a clean way to distinguish cross-lingual transfer from Turkish-side reaffirmation or relearning. Prior work either studies multilingual factual inconsistency without adaptation, adaptation without source attribution, or cross-lingual knowledge editing rather than broader target-language adaptation.

A further limitation is contamination. When naturally occurring real-world facts are used, it is often impossible to determine whether a model had already encountered those facts in multiple languages during pretraining or adaptation. This makes apparent target-language retrieval gains difficult to interpret: a correct Turkish answer may reflect genuine transfer, but it may also reflect prior multilingual exposure. A controlled synthetic-fact setup is therefore necessary to isolate the source of factual retrieval more reliably.

The main research gap, then, is the absence of a controlled experimental design that can separate transfer from Turkish-side reaffirmation and relearning during Turkish adaptation of a language model.

4 Research Question

To address this gap, this thesis investigates the following research question:

> **When factual knowledge becomes retrievable in Turkish after Turkish adaptation of a language model, does this primarily reflect cross-lingual transfer from previously acquired English knowledge, or reaffirmation and relearning from Turkish adaptation data?**

5 Proposed Methodology

This thesis proposes a controlled synthetic-fact framework designed to separate three effects:

1. cross-lingual transfer from previously acquired English knowledge,
2. Turkish-side reaffirmation or relearning through fact repetition during adaptation, and
3. genuinely new Turkish-side learning.

The study begins with a small or medium English-centric base model. A set of synthetic facts will be introduced in English only. The knowledge acquisition stage will first be implemented through continued pretraining on English data enriched with the target synthetic facts, optionally mixed with generic pretraining data to maintain model stability. If this does not reliably establish English-side retrieval of the target facts, a fallback design will train a smaller language model from scratch on standard pretraining data infused with the same synthetic facts. The use of fictitious facts follows the logic of controlled synthetic setups such as TOFU Maini et al., 2024, since synthetic knowledge makes it possible to track more precisely where and when a fact could have been acquired.

Synthetic fact generation

Synthetic facts will be constructed as controlled subject–relation–object triples and verbalized through short natural-language statements and question templates. The fact inventory will be designed to vary several properties that may affect later retrieval, especially fact frequency during English-side acquisition and the linguistic character of entity names. In particular, some entities will be given names that are more typical of English and others names that are more typical of Turkish. This makes it possible to examine not only whether transfer occurs, but also whether it depends on controlled properties of the training signal. To reduce contamination, the synthetic entities, relations, and combinations will be designed to avoid overlap with naturally occurring real-world facts.

Turkish adaptation branches

After English-side acquisition, the same synthetic-fact-enriched model will be adapted toward Turkish under multiple conditions:

- **Core Branch A (No Repetition):** Turkish adaptation without Turkish repetition of the target synthetic facts. This branch tests whether Turkish factual retrieval can emerge from adaptation alone, that is, without Turkish-side re-exposure to the facts.
- **Core Branch B (Repetition):** Turkish adaptation with Turkish repetition of some or all previously taught synthetic facts. This branch tests whether Turkish retrieval depends on reaffirmation or relearning from Turkish adaptation data.
- **Optional Branch C (New Turkish Facts):** Turkish adaptation with additional synthetic facts introduced only in Turkish. If resources permit, this branch will provide a further comparison between transfer of prior English knowledge and genuinely new Turkish-side learning.

This branching design enables a more informative comparison than a single adaptation condition. In particular, it allows the thesis to compare Turkish adaptation without fact repetition against Turkish adaptation with fact repetition, which is central to separating transfer from Turkish-side reaffirmation.

Probing strategy

The learned facts will be probed in both English and Turkish using parallel prompt templates derived from the same underlying synthetic triples. The main analysis will include only facts that are first shown to be retrievable in English before Turkish adaptation. After adaptation, the same facts will be queried in Turkish to test whether they have become accessible cross-lingually. In addition to direct prompt templates, the evaluation will include paraphrased variants where feasible in order to reduce the risk that results reflect only memorized surface forms rather than more robust factual access.

Illustrative example

As a simple example, consider the synthetic fact triple <*Leran Dovik, profession, river architect*>, where both the subject and the object are fictitious. During the English knowledge-acquisition stage, the model may be exposed to statements such as "Leran Dovik works as a river architect" or "The profession of Leran Dovik is river architect." The fact is then first verified through English probing, for example with the question "What is Leran Dovik's profession?" and the expected answer "river architect."

Afterward, the same fact is tested in Turkish under different adaptation conditions. In the no-repetition branch, Turkish adaptation data does not contain this fact, and the probing question may take the form “Leran Dovik’in mesleği nedir?” with the expected answer “nehir mimarı.” In the repetition branch, the Turkish adaptation data includes sentences such as “Leran Dovik nehir mimarı olarak çalışır,” which makes it possible to compare Turkish retrieval with and without Turkish-side reaffirmation. Where feasible, paraphrased probes such as “Leran Dovik hangi alanda çalışmaktadır?” will also be included to reduce dependence on a single surface form.

6 Evaluation Plan

The proposed setup will be evaluated through factual probing on the synthetic fact inventory in both English and Turkish. The evaluation is designed to measure not only whether the target facts are retrievable, but also under which adaptation conditions Turkish retrieval emerges and how this affects previously acquired English retrieval.

The primary comparison will track factual retrieval across languages and training stages:

- English retrieval before Turkish adaptation,
- Turkish retrieval before Turkish adaptation,
- English retrieval after Turkish adaptation, and
- Turkish retrieval after Turkish adaptation.

Only facts that are first shown to be retrievable in English before Turkish adaptation will be included in the main transfer analysis. This restriction ensures that the evaluation focuses on knowledge that was successfully acquired during the English-side knowledge acquisition stage.

The central analysis concerns the comparison between the Turkish adaptation branches. If Turkish retrieval improves in Branch A, where the target facts are not repeated in Turkish, this supports the interpretation that Turkish adaptation can enable cross-lingual access to previously acquired English knowledge. If retrieval improves substantially more in Branch B than in Branch A, this suggests that Turkish-side repetition contributes additional reaffirmation or relearning effects. If resources permit, Branch C will provide a further contrast between transfer of prior English knowledge and genuinely new Turkish-side learning.

In addition, English retrieval after Turkish adaptation will be measured to assess whether Turkish adaptation preserves or degrades previously acquired English factual knowledge.

This makes it possible to identify potential trade-offs between gains in Turkish retrieval and catastrophic forgetting on the English side.

The main outcome measures are as follows:

- **Accuracy:** exact factual retrieval on the synthetic fact set in English and Turkish;
- **Transfer gain:** the increase in Turkish retrieval for facts learned only in English and not repeated in Turkish adaptation data;
- **Reaffirmation gain:** the additional increase in Turkish retrieval when target facts are repeated in Turkish adaptation data;
- **English retention:** the preservation of English retrieval after Turkish adaptation;
- **Frequency-sensitive analysis:** whether facts taught with higher English-side frequency transfer more reliably than rare facts;
- **Name-sensitive analysis:** whether transfer differs between entities with more English-like and more Turkish-like names.

Where feasible, the evaluation will include both direct prompt templates and paraphrased probes in English and Turkish. This will help distinguish robust factual access from retrieval that depends only on a single memorized surface form.

7 Expected Contribution

This thesis is expected to make three main contributions. First, it proposes a controlled experimental design for studying whether target-language factual retrieval after Turkish adaptation reflects cross-lingual transfer or Turkish-side reaffirmation and relearning. Second, it provides empirical evidence on whether factual knowledge taught in English can become retrievable in Turkish without explicit repetition of the same facts during Turkish adaptation. Third, it offers a controlled analysis of how factors such as fact frequency and name distribution influence the emergence of cross-lingual factual retrieval.

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